

## SANGZHI, China

WMO: 575540

Lat: 29.4013N

Lon: 110.1622E

Elev: 322

StdP: 97.52

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |             |            |            | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD<br>to 99.6% DB |           | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|-------------|------------|------------|-----------------------|------------|------------|------------|--------------------------|-----------|--------------|
|               |             |             | 99.6%                         |            |            | 99%         |            |            | 0.4%                  |            | 1%         |            |                          |           |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP          | HR         | MCDB       | WS                    | MCDB       | WS         | MCDB       | MCWS                     | PCWD      |              |
| (a)<br>1      | (b)<br>-1.4 | (c)<br>-0.1 | (d)<br>-7.7                   | (e)<br>2.0 | (f)<br>4.6 | (g)<br>-5.6 | (h)<br>2.5 | (i)<br>3.4 | (j)<br>6.2            | (k)<br>6.5 | (l)<br>5.3 | (m)<br>5.3 | (n)<br>1.2               | (o)<br>40 | (p)<br>0.328 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|----------------------|------------|
|               |                        | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                      |            |
|               |                        | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                 | PCWD       |
| (a)<br>7      | (b)<br>8.0             | (c)<br>35.3     | (d)<br>25.6 | (e)<br>34.1 | (f)<br>25.2 | (g)<br>32.9 | (h)<br>24.7 | (i)<br>26.9         | (j)<br>32.4 | (k)<br>26.4 | (l)<br>31.6 | (m)<br>25.9 | (n)<br>30.7 | (o)<br>2.4           | (p)<br>200 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 25.6                            | 21.7 | 29.2 | 25.1 | 21.0 | 28.6 | 24.6 | 20.4 | 28.2 | 87.2          | 32.4 | 84.7 | 31.9 | 82.4 | 30.9 | 29.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 5.5           | (b) 4.6 | (c) 3.9 | (e) -2.6                   | (f) 37.4 | (g) 1.4            | (h) 1.0 | (i) -3.6                                           | (j) 38.1 | (k) -4.4   | (l) 38.7 | (m) -5.2   | (n) 39.3 | (o) -6.2   | (p) 40.0 |    |    |
| (a) 5.5           | (b) 4.6 | (c) 3.9 | (e) -3.2                   | (f) 27.8 | (g) 1.4            | (h) 0.7 | (i) -4.2                                           | (j) 28.3 | (k) -5.0   | (l) 28.8 | (m) -5.8   | (n) 29.2 | (o) -6.8   | (p) 29.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 17.2        | 5.3   | 7.4   | 12.0  | 17.4  | 21.6  | 24.8  | 27.7  | 27.5  | 23.5  | 18.0  | 12.7  | 7.4   |  |  |
|                                                                        | DBStd       | 8.31        | 3.09  | 3.54  | 3.99  | 3.74  | 2.86  | 2.59  | 2.25  | 2.58  | 3.10  | 3.13  | 3.31  | 2.63  |  |  |
|                                                                        | HDD10.0     | 358         | 147   | 86    | 25    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 13    | 86    |  |  |
|                                                                        | HDD18.3     | 1528        | 403   | 306   | 198   | 60    | 6     | 0     | 0     | 0     | 2     | 44    | 171   | 338   |  |  |
|                                                                        | CDD10.0     | 2977        | 2     | 14    | 87    | 223   | 359   | 445   | 550   | 541   | 406   | 248   | 93    | 7     |  |  |
|                                                                        | CDD18.3     | 1105        | 0     | 0     | 3     | 32    | 107   | 195   | 292   | 283   | 158   | 34    | 1     | 0     |  |  |
|                                                                        | CDH23.3     | 9798        | 0     | 1     | 32    | 256   | 714   | 1521  | 3057  | 2893  | 1142  | 176   | 8     | 0     |  |  |
|                                                                        | CDH26.7     | 3942        | 0     | 0     | 4     | 69    | 239   | 553   | 1337  | 1294  | 411   | 35    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 1.2         | 1.1   | 1.3   | 1.4   | 1.4   | 1.3   | 1.2   | 1.3   | 1.3   | 1.2   | 1.0   | 1.0   | 1.0   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 1392        | 26    | 37    | 77    | 145   | 201   | 245   | 195   | 153   | 112   | 107   | 65    | 30    |  |  |
|                                                                        | PrecMax     | 2229        | 64    | 87    | 180   | 265   | 382   | 695   | 507   | 434   | 261   | 305   | 163   | 90    |  |  |
|                                                                        | PrecMin     | 825         | 2     | 2     | 23    | 25    | 104   | 59    | 33    | 10    | 19    | 5     | 0     | 3     |  |  |
|                                                                        | PrecStd     | 299         | 15    | 21    | 43    | 57    | 74    | 133   | 129   | 97    | 56    | 62    | 42    | 20    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 16.8  | 20.7  | 26.7  | 31.3  | 33.2  | 34.7  | 36.7  | 37.0  | 35.1  | 30.0  | 24.4  | 17.8  |  |  |
|                                                                        |             | MCWB        | 9.7   | 12.8  | 16.8  | 19.7  | 20.9  | 24.5  | 26.4  | 25.4  | 24.7  | 20.2  | 16.4  | 11.0  |  |  |
|                                                                        | 2%          | DB          | 14.0  | 17.4  | 23.5  | 28.7  | 31.2  | 33.0  | 35.2  | 35.5  | 32.7  | 27.3  | 21.8  | 15.6  |  |  |
|                                                                        |             | MCWB        | 8.4   | 11.2  | 15.4  | 19.0  | 20.9  | 24.0  | 26.2  | 25.6  | 24.0  | 19.1  | 15.3  | 10.0  |  |  |
|                                                                        | 5%          | DB          | 11.7  | 14.8  | 20.9  | 26.4  | 29.4  | 31.6  | 34.0  | 34.2  | 30.9  | 25.3  | 19.5  | 13.5  |  |  |
|                                                                        |             | MCWB        | 7.6   | 9.9   | 14.2  | 18.3  | 21.0  | 24.0  | 25.8  | 25.4  | 22.9  | 18.5  | 14.5  | 9.1   |  |  |
|                                                                        | 10%         | DB          | 9.9   | 12.6  | 18.4  | 24.0  | 27.5  | 29.9  | 32.7  | 32.8  | 29.1  | 23.1  | 17.6  | 11.7  |  |  |
|                                                                        |             | MCWB        | 6.9   | 9.2   | 13.2  | 17.5  | 20.5  | 23.6  | 25.6  | 25.0  | 22.0  | 17.9  | 13.8  | 8.3   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 11.4  | 14.4  | 19.0  | 22.8  | 24.9  | 26.9  | 27.7  | 27.7  | 26.2  | 22.9  | 18.1  | 12.7  |  |  |
|                                                                        |             | MCDB        | 14.1  | 18.4  | 22.8  | 27.7  | 29.1  | 30.5  | 33.3  | 33.4  | 32.2  | 26.6  | 21.0  | 15.3  |  |  |
|                                                                        | 2%          | WB          | 9.4   | 12.1  | 16.6  | 21.0  | 23.4  | 26.0  | 27.0  | 26.9  | 25.3  | 20.7  | 16.6  | 11.0  |  |  |
|                                                                        |             | MCDB        | 12.1  | 15.4  | 21.0  | 25.3  | 27.0  | 30.1  | 32.6  | 32.7  | 30.6  | 24.3  | 19.8  | 14.1  |  |  |
|                                                                        | 5%          | WB          | 8.3   | 10.8  | 15.1  | 19.4  | 22.5  | 25.3  | 26.5  | 26.2  | 24.5  | 19.7  | 15.4  | 9.7   |  |  |
|                                                                        |             | MCDB        | 11.0  | 13.4  | 19.0  | 23.4  | 26.2  | 29.2  | 31.8  | 31.5  | 29.0  | 22.9  | 18.4  | 12.6  |  |  |
|                                                                        | 10%         | WB          | 7.1   | 9.7   | 13.8  | 18.3  | 21.5  | 24.6  | 26.1  | 25.7  | 23.6  | 18.7  | 14.4  | 8.6   |  |  |
|                                                                        |             | MCDB        | 9.6   | 12.3  | 17.5  | 22.5  | 25.2  | 28.2  | 31.1  | 30.6  | 27.2  | 22.1  | 16.9  | 11.2  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.8   | 6.5   | 7.8   | 8.5   | 8.2   | 7.5   | 8.0   | 8.5   | 8.0   | 7.1   | 6.8   | 6.4   |  |  |
|                                                                        |             | MCDBR       | 11.4  | 12.4  | 13.9  | 14.1  | 13.2  | 11.0  | 10.5  | 11.0  | 11.2  | 11.8  | 11.2  | 11.4  |  |  |
|                                                                        | 5% WB       | MCWBR       | 6.3   | 6.4   | 6.0   | 5.1   | 4.3   | 3.5   | 2.7   | 2.7   | 3.4   | 4.2   | 5.1   | 6.2   |  |  |
|                                                                        |             | MCWBR       | 9.4   | 9.7   | 11.1  | 10.3  | 8.8   | 8.1   | 9.1   | 9.6   | 9.2   | 8.6   | 8.8   | 9.9   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | taub        | 0.693 | 0.735 | 0.766 | 0.747 | 0.668 | 0.662 | 0.606 | 0.667 | 0.696 | 0.721 | 0.702 | 0.677 |  |  |
|                                                                        |             | taud        | 1.460 | 1.423 | 1.410 | 1.485 | 1.665 | 1.709 | 1.856 | 1.696 | 1.611 | 1.503 | 1.502 | 1.496 |  |  |
|                                                                        | Ebn at Noon | Ebn at Noon | 529   | 560   | 585   | 623   | 680   | 681   | 718   | 669   | 626   | 561   | 520   | 513   |  |  |
|                                                                        |             | Edh at Noon | 260   | 292   | 314   | 299   | 251   | 239   | 206   | 239   | 253   | 265   | 247   | 241   |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | RadAvg      | 1.59  | 1.84  | 2.73  | 3.35  | 3.59  | 3.81  | 4.60  | 4.34  | 3.34  | 2.50  | 2.05  | 1.64  |  |  |
|                                                                        | RadStd      | RadStd      | 0.33  | 0.44  | 0.27  | 0.46  | 0.63  | 0.58  | 0.53  | 0.43  | 0.45  | 0.41  | 0.39  | 0.34  |  |  |

## Historical Trends

|                             |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                             |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                             |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only           |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (6 neighbors) |  | N/A   | N/A     | N/A    | +0.52   | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page